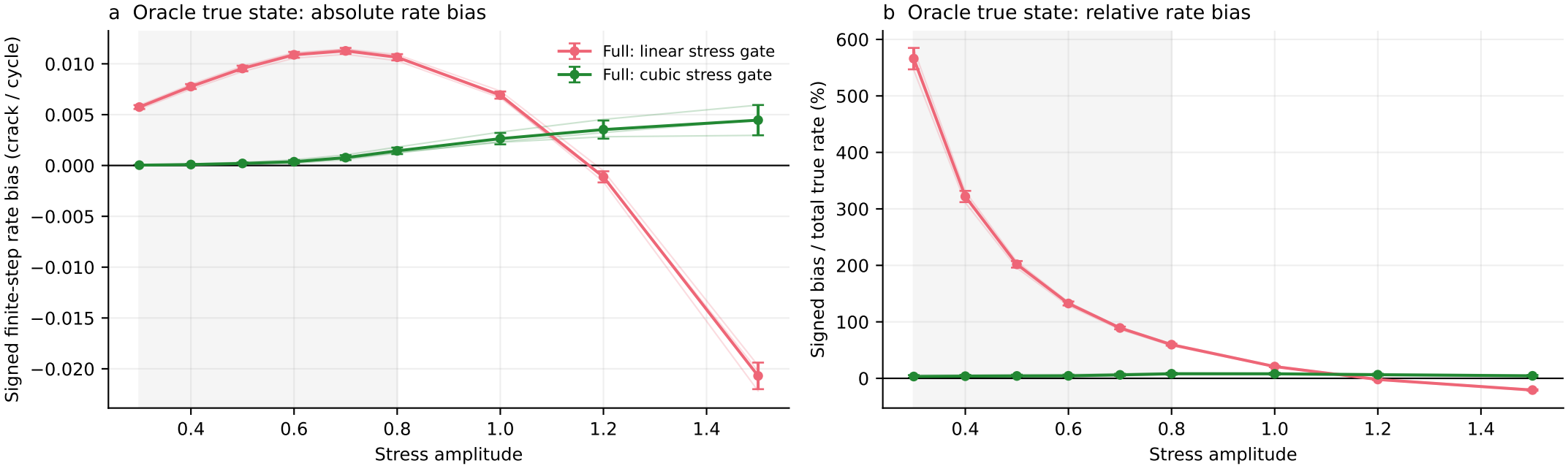
# Does the learned update overestimate low-load damage?




Thin lines: individual seeds; thick lines and bars: mean +/- sample SD across 3 seeds.
Fixed (initial crack, current crack, stress, coefficient, cycles) grid; true crossing steps excluded.
Shading marks low-load range. Finite-step rates use uncapped updates; this is a synthetic-material diagnostic.